

PC Configuration

Desktop → IP Configuration

Setting	Value
IP Address	192.168.1.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

Router Configuration (SSH)

```
enable
configure terminal

hostname Router1
ip domain-name cnslab.com

interface gig0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

username CNSLab privilege 15 secret cisco

service password-encryption

crypto key generate rsa
1024

ip ssh version 2

line vty 0 4
login local
transport input ssh
exit

copy running-config startup-config
```

Switch Configuration (SSH)

```
enable
configure terminal

hostname Switch1
ip domain-name cnslab.com

interface vlan 1
ip address 192.168.1.10 255.255.255.0
no shutdown
```

```
exit

ip default-gateway 192.168.1.1

username CNSLab privilege 15 secret cisco

service password-encryption

crypto key generate rsa
1024

ip ssh version 2

line vty 0 4
login local
transport input ssh
exit

copy running-config startup-config
```

SSH Testing from PC

Desktop → Command Prompt

```
ping 192.168.1.1
ssh -l CNSLab 192.168.1.1
```

Password:

```
cisco
```

Telnet Configuration (If Asked Instead of SSH)

```
enable
configure terminal

interface gig0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

username CNSLab password cisco

line vty 0 4
login local
transport input telnet
exit

copy running-config startup-config
```

AAA Authentication using RADIUS + SSH (Final Lab Steps)

1. ROUTER CONFIGURATION

```
enable

configure terminal

hostname R1

interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

aaa new-model

radius-server host 192.168.1.2 key cisco

aaa authentication login default group radius local

username backup privilege 15 secret backup123

ip domain-name lab.local

crypto key generate rsa
1024

ip ssh version 2

line vty 0 4
transport input ssh
login authentication default
exit

end

write memory
```

2. SERVER CONFIGURATION

Desktop → IP Configuration

IP Address: 192.168.1.2
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

Services → AAA

AAA Service: ON

Add Router Client

Client Name: R1
Client IP: 192.168.1.1
Secret: cisco
Server Type: Radius

Click:

Add

Add User

Username: admin
Password: cisco123

Click:

Add

3. PC CONFIGURATION

Desktop → IP Configuration

IP Address: 192.168.1.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

4. TESTING

PC → Command Prompt

Ping Test

ping 192.168.1.1
ping 192.168.1.2

SSH Login Test

ssh -l admin 192.168.1.1

Password:

cisco123

CNS Experiment 3 — Final Practical Steps

Topology Connections

- PC1 Fa0 → Switch Fa0/1
- PC2 Fa0 → Switch Fa0/2
- PC3 Fa0 → Switch Fa0/3
- Switch Fa0/5 → Router G0/0
- Router G0/1 → Server Fa0

PC1 Configuration

Desktop → IP Configuration

IP Address: 192.168.1.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

PC2 Configuration

Desktop → IP Configuration

IP Address: 192.168.1.20
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

PC3 Configuration

Desktop → IP Configuration

IP Address: 192.168.1.30
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

Server Configuration

Desktop → IP Configuration

IP Address: 192.168.2.100
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.2.1

Router Configuration

CLI Commands

```
enable
configure terminal

interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface g0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
```

Connectivity Test

PC1 → Desktop → Command Prompt

ping 192.168.2.100

Standard ACL Configuration

Router CLI

```
enable
configure terminal

access-list 10 deny 192.168.1.20 0.0.0.0
access-list 10 permit any

interface g0/0
ip access-group 10 in
exit
```

Standard ACL Testing

From PC1

ping 192.168.2.100

From PC2

ping 192.168.2.100

Extended ACL Configuration

Router CLI

```
enable
configure terminal

access-list 100 permit tcp host 192.168.1.10 host
192.168.2.100 eq 80
access-list 100 deny tcp host 192.168.1.20 host
192.168.2.100 eq 21
access-list 100 permit ip any any

interface g0/0
no ip access-group 10 in
ip access-group 100 in
exit
```

Enable Server Services

Server → Services

HTTP

HTTP → ON

FTP

FTP → ON

HTTP Test

PC1 → Desktop → Web Browser

http://192.168.2.100

Verification Commands

Router CLI

```
show access-lists
show ip interface
show running-config
```